# What was added

Two new capabilities, explained without jargon

ARBITER · Pfizer Digital & Technology Hackathon 2026 · Team BU 1 — Jack He, Andres Lopez, Jose Cruz-Lopez
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> Two things were built. The first lets an organisation run ARBITER on **its own compound** instead of only on our public test set. The second is the groundwork for **measuring** the claim our whole pitch rests on — that asking an AI directly does not give you a decision you can defend.

## At a glance

| WHAT | WHAT IT MEANS IN THE ROOM | STATE |
|---|---|---|
| **Intake** | "Can I run this on *my* drug?" — the answer is now yes, live, in the browser | Working and demonstrable |
| **AI comparison** | Turns "an AI would give you a different answer each time" from a claim into a measurement | Built and tested, not yet run |
| **Documentation** | Anyone new can now understand the project without a walkthrough | Complete |

## Feature 1 — Run it on your own compound

### Intake

WORKING, DEMONSTRABLE

WHAT IT DOES

A new screen where a scientist enters the studies they have on their own compound — what was tested, what it found, in what system, at what dose, and how reliable the study was. ARBITER then reasons over that evidence exactly as it does over ours, and produces a position with the full argument behind it.

THE INSIGHT THAT SHAPED IT

**You do not upload a drug. You upload evidence.** If you give the system only a molecule, it has exactly one weak piece of evidence to go on and will correctly refuse to decide — every single time. That would look broken. So the screen asks for the studies, which is what an organisation actually has and what the public data set is missing.

WHY THIS IS COMMERCIALLY THE IMPORTANT ONE

Our published result is that ARBITER declines to commit on 97% of compounds, because public data is thin. **An organisation's own data is not thin.** They hold the cell studies, the transporter assays, and above all the human dose data whose absence causes most of those declines. So ARBITER should perform noticeably *better* inside a company than it does on our benchmark — and this feature is what lets someone see that for themselves.

### The part worth demonstrating live

The screen tells you **in advance** whether your evidence could ever produce a decision. Enter a weak, computer-predicted study and it says:

> *On this evidence no verdict is reachable at any confidence values — the most it could muster is 0.150 against the 0.500 a verdict requires. Committing would need evidence from a human system; a study that measures an actual biological mechanism; and a finding at a dose a patient really receives.*

Add a human study measuring a real mechanism at a realistic dose, and it flips to "*this evidence can reach a committed position.*" Create the compound and it opens with a full verdict, the argument trace, and the single change that would reverse it.

**That is the honesty of the product made visible.** Most tools give you a confident number and let you find out later it was unfounded. This one tells you what your evidence can and cannot support before you rely on it.

> **Three protections built in, worth mentioning if challenged**
>
> **Your compound cannot contaminate our published results.** Anything entered is kept separate from the benchmark, enforced in the code rather than by policy, and it is refused outright if its identifier collides with a compound already in our test set.
> **Nothing leaves the browser.** No server, no upload, no account. It disappears on reload.
> **The rules cannot be bent to suit your compound.** They stay locked and fingerprinted, exactly as they are for the benchmark.

## Feature 2 — Measuring what an AI actually does here

### The AI comparison

BUILT AND TESTED, NOT YET RUN

THE QUESTION IT ANSWERS

*"Why not just ask ChatGPT?"* — the question most likely to be asked, and the one our pitch depends on answering well.

THE EXPERIMENT

Give a top-tier AI the same evidence and the same rules ARBITER uses, ask it the same question twenty-five times, and count how often it contradicts itself. ARBITER, asked a thousand times, returns an identical answer every time — that part is already proven.

WHAT WAS BUILT

Everything except the spending. The scoring, the exact wording of the question, the record-keeping that lets a reviewer re-derive our number, and the handling for a model that declines to answer a medical question at all. Thirty-eight automated checks, all passing. **No AI account and no money were needed for any of it**, and none has been spent.

WHAT IT STILL NEEDS

An AI account and roughly $27, plus a decision about which provider. Then a small trial run before committing the full budget.

**One measurement worth mentioning**

A subtle trap was found and closed. If an AI is simply guessing at random between two answers, it will still *appear* to agree with itself about **58% of the time** — purely by chance. Anyone reading a score of 0.6 as "fairly consistent" would be reading noise as a result.

So that floor is now calculated exactly and will be printed next to our figure. It also corrected a number in our own design document, which had estimated 0.580 by simulation where the true value is 0.5806. **The difference changes nothing — but we wrote it down rather than quietly fixing it**, which is the same discipline that makes the rest of our numbers trustworthy.

## Supporting work

The project had a one-line README. Anyone new — a judge, a teammate, a colleague inheriting this — had no way in short of a personal walkthrough. It now has a full front door explaining the goal, how the system works, the honest results including the unflattering ones, and where to read next.

The internal handover document was also brought up to date and corrected in three places where it had drifted from what the software actually does.

## By the numbers

| MEASURE | BEFORE | AFTER |
|---|---|---|
| Automated tests passing | 552 | **623** |
| Screens in the app | 6 | **7** |
| Lines of code and documentation added | — | **~2,500** |
| Published results changed | — | **none** |

That last row is deliberate and worth saying aloud: **none of this touched a reported number.** Everything we have published stands exactly as it did, which is checked automatically on every change.

## What is honestly not finished

- **The AI comparison has never been run.** It needs an account, about $27, and a decision on which provider. Until then our central claim is an argument, not a measurement.
- **Evidence must be typed in by hand.** Uploading a spreadsheet, and having AI read a study report and fill the form in for you, are both designed but not built.
- **A custom compound gets no computer-generated prediction.** That model cannot score a molecule it has never seen. It costs almost nothing — the system heavily discounts that kind of evidence anyway — and the screen now says so plainly rather than leaving it silently absent.

None of this sits on the critical path to submission, and the new screen is one line of code away from being removed if the schedule needs it.

### Three sentences for the meeting

**1.** "You can now run ARBITER on your own compound, in the browser, with nothing leaving your machine — and it tells you up front whether your evidence is even capable of supporting a decision."

**2.** "Our published result looks weak because public data is thin. An organisation's own data is not, and this is the feature that lets someone prove that to themselves."

**3.** "We also built everything needed to measure how inconsistently an AI answers this same question. It costs about $27 to run, and it turns our central argument into evidence."